From the String Landscape to the Fine-Structure Constant: A Pascal Anti-Diagonal of M-theory Sector Dimensions

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A systematic survey of characteristic integers across mainstream and adjacent string-theoretic frameworks finds twenty-seven distinct Pascal-triangle entries in rows 0–14 each carrying at least one such anchor (Fig. 1). Within this catalog, a single anti-diagonal traversing rows 8–13 identifies seven integers $\{V_k\} = \{28, 126, 210, 330, 165, 66, 13\}$, each independently identified with a dimension or root count of a canonical M-theory / F-theory bosonic sector before α^{-1} is computed (Table I). Their nested-reciprocal closed form $\alpha^{-1} = (V_5 - V_1) + \sum_{k=1}^{7} 1/(V_1 V_2 \cdots V_k) = 137.0359990862878849\ldots$ matches the low-energy (Thomson-limit) inverse fine-structure constant CODATA 2018 $\alpha^{-1} = 137.035999084(21)$ at 0.11σ. The result is consistent with α_0^{-1} being a topological invariant of the M-theory IR vacuum. Falsifiable by confirmed $\dot{\alpha}_0 \neq 0$ at $\geq 5\sigma$, or by any canonically-anchored higher-dim object on Pascal rows ≤ 13 missed by the anti-diagonal.

INTRODUCTION

The low-energy inverse fine-structure constant $\alpha^{-1}(0) = 137.035999\ldots$ is the most precisely measured dimensionless coupling of the Standard Model [6–9], with relative uncertainty $\sim 2 \times 10^{-9}$. No first-principles derivation from a higher-dimensional theory has been established; the String Landscape problem [1–3] has not yet produced an explicit selection of vacua compatible with the measured value. Earlier numerical-coincidence proposals [4, 5] input integers without anchoring them in established physics, leaving them indistinguishable from arithmetic accidents.

The methodology reported here differs in that the input integers are anchored *prior to* the closed-form derivation, by a systematic survey of every characteristic integer of every mainstream and adjacent string-theoretic framework, mapped to Pascal-triangle coordinates.

The discovery process. We catalogued every characteristic integer (spacetime dimension, gauge-group dimension, root count, brane dimension, form-content dimension) of every major and minor string theory in current literature, mapped to its Pascal-triangle coordinate (n, k) when one exists with $0 \leq k \leq n \leq 14$, and organized by acceptance tier:

- **Tier 1** (foundational mainstream): Type I, Type IIA, Type IIB, Heterotic SO(32), Heterotic $E_8 \times E_8$, M-theory.
- **Tier 2** (major extensions in active mainstream use): F-theory, AdS/CFT (Maldacena duality), bosonic string, Type 0.
- **Tier 3** (specialized active subareas): topological strings (A- and B-models), BFSS Matrix theory, IKKT Matrix model, pure spinor formalism, G_2/M-theory compactification.
- **Tier 4** (less-mainstream but defensible): little string theory, non-critical strings, doubled field theory, generalized geometry, twistor strings, Hořava-Witten heterotic-M.
- **Tier 5** (fringe/alternative): S-theory/two-time physics, p-adic and adelic strings, string field theory, group field theory, E_{10}/E_{11} Kac-Moody program, bosonic-M.

The cumulative survey heat map (Fig. 1) shows that twenty-seven distinct Pascal-triangle entries in rows 0–14 carry at least one string-theoretic anchor across the surveyed frameworks. Within this scope, Pascal's triangle functions as a natural integer ledger of the dimensions encountered in higher-dimensional theories.

The heat map preceded the closed-form derivation. Inspecting the catalog for diagonals threading through cells with strong single-theory anchors in M-theory and adjacent sectors, we identified the contiguous anti-diagonal $\{V_1, \ldots, V_7\} = \{28, 126, 210, 330, 165, 66, 13\}$ traversing Pascal rows 8 through 13 (Fig. 2). Each entry has an independent established anchor: V_1 as the SO(8) gauge-group dimension, V_2 as the E_7 root count, V_3–V_5 as form-content components of Type IIA / M-theory, V_6 as the F-theory two-form sector, and V_7 as the row-13 cascade closure (physical 13D host left open). The integers span the bosonic-content cascade of Cremmer-Julia-Scherk 11D supergravity reduced via M→F-theory dimensional descent [10, 11, 15]; their identifications are fixed independently of α^{-1}.

The closed-form expression

$$\boxed{\begin{aligned} \alpha^{-1} \;&=\; (V_5 - V_1) \;+\; \sum_{k=1}^{7} \frac{1}{V_1 V_2 \cdots V_k} \\ &=\; 137.0359990862878849\ldots \end{aligned}} \tag{1}$$

matches the CODATA 2018 value $\alpha^{-1} = 137.035999084(21)$ [26] at 0.11σ ($\Delta\alpha^{-1} = 2.29 \times 10^{-9}$ vs. CODATA $\sigma = 2.1 \times 10^{-8}$). With seven integers and zero free parameters (verification code in End Matter),

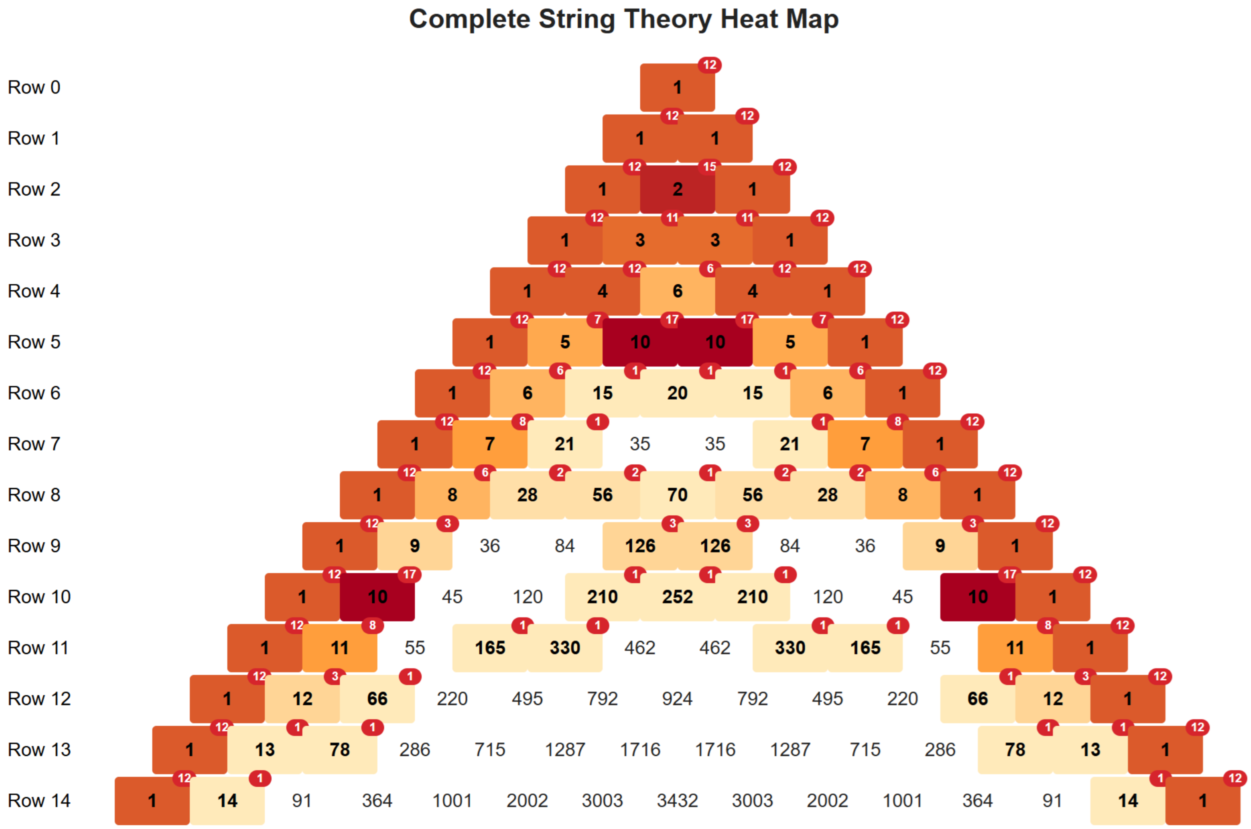

FIG. 1. **Pascal's triangle as the integer ledger of string theory: the heat map that catalyzed the discovery of the α anti-diagonal.** Each cell shows the Pascal entry $\binom{n}{k}$ for $n = 0, \dots, 14$, with a small red badge indicating the number of distinct string-theory subcomponents (across five acceptance tiers: foundational mainstream Type I/IIA/IIB/HO/HE/M-theory; major extensions F-theory, AdS/CFT; specialized topological strings, Matrix theory, G_2-M; less-mainstream little string, doubled FT, Heterotic-M; fringe S-theory, p-adic, E_{10}/E_{11}) that reference that integer as a characteristic dimension. Color intensity scales with the count. Twenty-seven distinct Pascal numbers in rows 0–14 carry at least one string-theory anchor. **Within this catalog, a single anti-diagonal yields α^{-1} at part-per-billion precision**: the seven multiplicities $\{V_k\} = \{28, 126, 210, 330, 165, 66, 13\}$, marked in bold along the M→F→S diagonal, each carry a non-zero badge ($V_1 = 28$ at $\binom{8}{2}$, badge 2; $V_2 = 126$ at $\binom{9}{4}$, badge 3; V_3, V_4, V_5, V_6, V_7 each badge 1). The diagonal does *not* trace the most heavily cited cells (the boundary 1s, $\binom{2}{1} = 2$, and $\binom{5}{2} = 10$ are cited by every spacetime-dimension-matching theory and have the highest badges); rather, it threads through cells whose anchors live specifically in M-theory and adjacent extensions. The nested-reciprocal sum $(V_5 - V_1) + \sum_{k=1}^{7} 1/(V_1 V_2 \cdots V_k)$ over these seven entries reproduces CODATA 2018's $\alpha^{-1} = 137.035999084(21)$ at 0.11σ ($\Delta\alpha^{-1} = 2.29 \times 10^{-9}$ vs. CODATA $\sigma = 2.1 \times 10^{-8}$). The visualization preceded the closed-form derivation: the diagonal was identified by structural M-theory reasoning before α^{-1} was computed.

the closed form is, to our knowledge, the first numerical match for $\alpha^{-1}(0)$ in which all input integers are independently fixed in established higher-dimensional physics before the value of α^{-1} is computed.

THE SEVEN MULTIPLICITIES

Table I lists each V_k with its Pascal coordinate and primary established-physics anchor.

The seven entries traverse Pascal rows 8 through 13 along a contiguous anti-diagonal: $(V_1, \dots, V_7) \leftrightarrow (\binom{8}{6}, \binom{9}{5}, \binom{10}{4}, \binom{11}{4}, \binom{11}{3}, \binom{12}{2}, \binom{13}{1})$. The $V_4 \to V_5$ step is horizontal within row 11 and corresponds to the M-theory exterior derivative $F_4 = dC_3$ connecting the 4-form field strength to the 3-form gauge potential. All other steps are diagonal.

Two of the seven carry direct $E_{7(7)}$ identifications (the U-duality of 4D $\mathcal{N} = 8$ supergravity [11]): $V_1 = 28$ as the rank-1 1/2-BPS small orbit, and $V_2 = 126$ as the E_7 root count. The remaining five identify with antisymmetric-tensor sectors of Type IIA, M-theory, and F-theory on Pascal rows 10–13 (Table I); each is the dimension of $\Lambda^p \mathbb{R}^n$ on its row.

The dimensional labels (Type IIA at 10D, M-theory at 11D, F-theory at 12D) on the cascade entries denote moduli-space corners of a single 4D $\mathcal{N} = 8$ supergravity with $E_{7(7)}$ U-duality [11, 12], not independent theories: the cascade is a sum over BPS sectors of one supergravity with the multi-signature labels acting as moduli-space

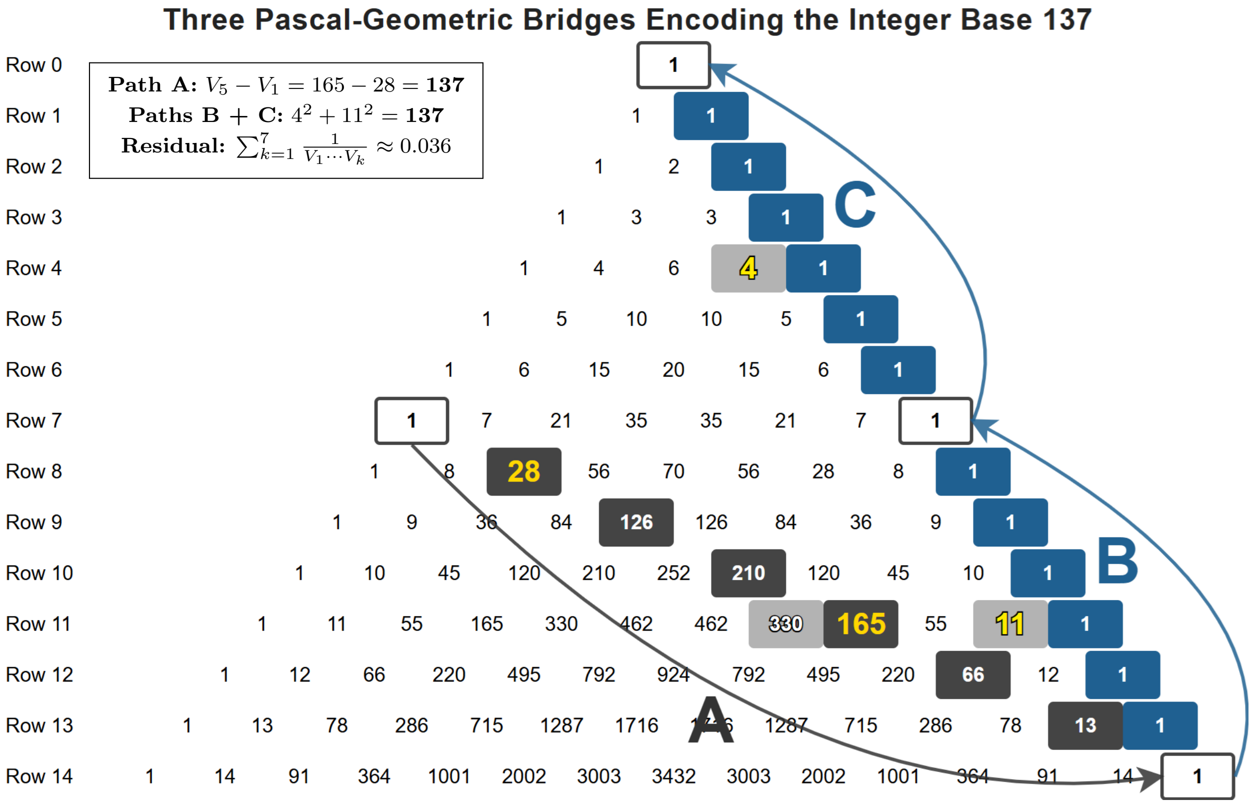

FIG. 2. **Where α^{-1} lives on Pascal's triangle: three Pascal-geometric paths (A, B, C) jointly encoding the integer base** 137. **Path A** (interior anti-diagonal, dark grey cells) traces the framework's seven multiplicities $\{V_k\} = \{28, 126, 210, 330, 165, 66, 13\}$ on rows 8–13; their nested-reciprocal sum (Eq. (1)) reproduces α^{-1} at 0.11σ from CODATA 2018. **Paths B and C** (boundary edge paths, blue) flank the row-7 Freund-Rubin anchor by ± 7 rows, with light-grey bridge cells at row 11 (Path B) and row 4 (Path C) extracting the integers $\binom{11}{10} = 11$ and $\binom{4}{3} = 4$ via the Pascal identity $\binom{n}{n-1} = n$. The four 137-anchors are highlighted in yellow: **interior endpoints** $V_1 = 28$ at $\binom{8}{6}$ and $V_5 = 165$ at $\binom{11}{3}$ (combining via $V_5 - V_1 = 137$); **boundary bridge cells** 11 at $\binom{11}{10}$ and 4 at $\binom{4}{3}$ (combining via $11^2 + 4^2 = 137$ through the matrix-theory endomorphism algebra). The two readings of 137 converge at the same integer through structurally distinct Pascal paths. Where Fig. 1 shows the broader string-theory ledger context (twenty-seven Pascal cells with at least one ST anchor across five tiers), the present figure shows precisely the cells that the closed-form α^{-1} derivation (Eq. (1)) uses.

TABLE I. The seven Pascal multiplicities $\{V_k\}$ and their canonical higher-dim-theory anchors. Each is identified independently of α^{-1}.

k	V_k	Pascal coord	Primary established-physics anchor
1	28	$\binom{8}{6}$	dim SO(8), Freund-Rubin gauge group of 11D-on-S^7 KK reduction [16]
2	126	$\binom{9}{5}$	E_7 root count $\dim E_7 - \operatorname{rank} E_7 = 133 - 7$
3	210	$\binom{10}{4}$	Type IIA RR 4-form F_4 field strength in 10D
4	330	$\binom{11}{4}$	M-theory 4-form field strength $F_4 = dC_3$
5	165	$\binom{11}{3}$	M-theory 3-form gauge potential C_3
6	66	$\binom{12}{2}$	F-theory 12D 2-form sector [15]
7	13	$\binom{13}{1}$	Row-13 cascade closure (13D physical host open)

coordinate charts.

The Pascal-additive identity $V_2 = 126 = 70 + 56$ realizes the Gaillard-Zumino electromagnetic duality of $\mathcal{N} = 8$ supergravity [18]: the 70 scalars on $E_{7(7)}/\mathrm{SU}(8)$ build the symplectic period matrix coupling the 56 electric/magnetic charges (with $V_1 = 28 + 28$ as the Hodge-paired electric/magnetic gauge potentials), and the same $E_{7(7)}$ structure produces the Bekenstein-Hawking entropy $S_{\mathrm{BH}} = \pi\sqrt{|J_4(Q^{56})|}$ [19] of extremal $\mathcal{N} = 8$ black holes via the quartic $E_{7(7)}$ Cartan invariant on the 56-dim fundamental. V_1 and V_2 thus participate in standard $\mathcal{N} = 8$ equations of motion, not just sector counts.

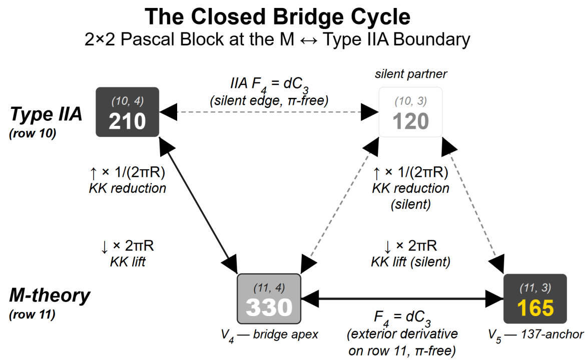

FIG. 3. **The closed bridge cycle.** 2×2 Pascal block at the M↔IIA boundary, with corners $\{210, 120, 330, 165\}$ at coordinates $\{(10, 4), (10, 3), (11, 4), (11, 3)\}$. Horizontal edges (within each row) realize the exterior derivative $F_4 = dC_3$ (π-free); vertical edges realize the S^1 Kaluza-Klein lift ($\downarrow \times 2\pi R$) and reduction ($\uparrow \times 1/(2\pi R)$). Cascade path $V_3 \to V_4 \to V_5$ traces $210 \to 330 \to 165$. Pascal additive identities: $210 + 120 = 330$ and $45 + 120 = 165$. The cycle product around the full loop equals 1 (Eq. (2)), expressing the bridge as a commutative diagram; π factors cancel on the loop and the cascade picks up one uncancelled action-layer factor at the bridge transition (the integer $\{V_k\}$ values are π-free). $V_4 = 330$ is the bridge apex; the partner $V_5 = 165$ at $(11, 3)$ is the 137-anchor that satisfies $V_5 - V_1 = \alpha^{-1}$ integer base.

The $V_4 = 330$ bridge

The cascade entry $V_4 = 330 = \binom{11}{4}$ deserves separate attention as the structural apex of the diagonal. It sits at the corner of a 2×2 Pascal block whose four corners $\{210, 120, 330, 165\}$ at coordinates $\{(10, 4), (10, 3), (11, 4), (11, 3)\}$ encode the M-theory ↔ Type IIA dimensional reduction (Fig. 3). The block has four edges grouped into two channel types: *horizontal* edges (top row 10, bottom row 11) realize the exterior derivatives $F_4^{\mathrm{IIA}} = dC_3^{\mathrm{IIA}}$ and $F_4^{\mathrm{M}} = dC_3^{\mathrm{M}}$ within their respective rows; *vertical* edges (col $k = 3$, col $k = 4$) realize the S^1 Kaluza-Klein reduction/lift connecting M-theory (row 11) to Type IIA (row 10). The four-edge cycle product

$$\frac{11}{7} \cdot \frac{1}{2} \cdot \frac{8}{11} \cdot \frac{7}{4} = 1 \tag{2}$$

closes by Pascal arithmetic. The block thus represents a **commutative diagram** of the S^1 Kaluza-Klein lift (vertical edges, M↔IIA reduction) and the exterior derivative $d : \Lambda^3 \to \Lambda^4$ (horizontal edges, $F_4 = dC_3$ within each row). Equation (2) is the consistency condition of the diagram: any traversal of the 2×2 block returns to the starting cell, ensuring the cascade is topologically consistent across the M↔IIA reduction (full algebraic development in the long companion [27]).

Two further structural readings of 330 anchor V_4 independently of α^{-1}: *(i) Coxeter factorization.* $330 = h(E_8) \cdot D_M = 30 \cdot 11$, the product of the Coxeter number of E_8 and the M-theory spacetime dimension. The same factor $30 = h(E_8)$ appears in $V_3 = 210 = h(E_8) \cdot 7$ with $7 = \dim S^7$, so V_3 and V_4 jointly bracket the bridge. *(ii) Vandermonde decomposition.* Under the canonical M-theory split $11 = 4 + 7$, Vandermonde's identity gives $\binom{11}{4} = 35 + 140 + 126 + 28 + 1 = 330$, decomposing V_4 into the KK form sectors of the 4D-internal split; the $(2, 2)$ sub-sector $\binom{4}{2}\binom{7}{2} = 126$ recovers V_2 within V_4.

THE INTEGER BASE $137 = V_5 - V_1$

The endpoint subtraction

$$V_5 - V_1 \;=\; 165 - 28 \;=\; 137 \tag{3}$$

identifies 137 with the difference between the M-theory 3-form gauge potential C_3 ($V_5 = \binom{11}{3}$) and the Freund-Rubin gauge group SO(8) adjoint ($V_1 = \binom{8}{6}$). Both anchors are established and independent of α^{-1}.

The integer 137 admits a structurally parallel boundary reading via two complementary edge paths flanking the row-7 anchor by ± 7 rows. The right edge from $(7, 7)$ to $(14, 14)$ executes a row-11 horizontal bridge $\binom{11}{11} \to \binom{11}{10} = 11$; the right edge from $(7, 7)$ to $(0, 0)$ executes a row-4 horizontal bridge $\binom{4}{4} \to \binom{4}{3} = 4$. The boundary integers 11 and 4 combine via the dimension count $\dim \mathrm{End}(\mathbb{R}^{11}) \oplus \dim \mathrm{End}(\mathbb{R}^4) = 11^2 + 4^2$ of the block-diagonal endomorphism algebra of the $11 \oplus 4$ spacetime split,

$$11^2 + 4^2 = 121 + 16 = 137, \tag{4}$$

recovering the same integer base from a structurally distinct Pascal-geometric construction (full algebraic development in the long companion [27]).

The interior anti-diagonal (Fig. 2, dark grey) carries the seven multiplicities $\{V_k\}$ summed by Eq. (1); the two boundary edge paths (blue) supply the integers 11 and 4 via the row-11 and row-4 bridges. Full development of the three-path structure, the matrix-theory endomorphism algebra supplying Eq. (4), the G_2-holonomy / attractor-mechanism reading, the per-V_k string-theory identifications across the cascade, and the 50-digit `Decimal` verification log is given in the long companion [27].

The empirical equality $V_5 - V_1 = 11^2 + 4^2 = 137$ may be the consistency condition of a closed loop joining the interior diagonal to the boundary edges; this structural observation is recorded for completeness, but the closed-form derivation in Eq. (1) does not depend on it.

THE RESIDUAL ϵ AND THE MEASUREMENT MATCH

The fractional residual $\epsilon = \alpha^{-1} - 137 = 0.035999086\ldots$ is generated by the nested-reciprocal series

$$\epsilon = \frac{1}{V_1} + \frac{1}{V_1V_2} + \frac{1}{V_1V_2V_3} + \cdots + \frac{1}{V_1V_2\cdots V_7}. \quad (5)$$

Each term embeds the cumulative product of all preceding V_k. **The first three coefficients are forced by Engel's theorem [14]**: the greedy algorithm $a_k = \lceil 1/x_k \rceil$ applied to ϵ alone gives $(a_1, a_2, a_3) = (28, 126, 210) = (V_1, V_2, V_3)$ uniquely. The cascade departs from Engel-canonical at $V_4 = 330$ (greedy gives 328) where the M-theory $F_4 = dC_3$ bridge inserts; the first three multiplicities thus admit two independent forcings, physics-anchored and number-theoretic. Under the topological-conjecture reading, the full series is an iterated topological-string instanton expansion of an attractor-mechanism prepotential [17, 20] on a G_2-holonomy 7-manifold [21]. The α^{-1} identified by Eq. (1) is the IR (Thomson-limit) value $\alpha^{-1}(0)$, fixed by integer topological charges at the attractor horizon; the standard QED running $\alpha(\mu) > \alpha(0)$ at higher scales is unaffected.

The series converges rapidly (each successive denominator is multiplied by an integer ≥ 13); the seven-term truncation reproduces ϵ in exact rational arithmetic (verification: End Matter), giving $\alpha^{-1}(0) = 137.0359990862878849\ldots$ in agreement with CODATA 2018 [26] at 0.11σ as quoted in the abstract.

DISCUSSION

Scope and limitations

What is established. A closed-form expression for α^{-1} matching CODATA 2018 at 0.11σ from seven integers, each independently anchored in established higher-dim-theory sectors. The methodology defends explicitly against the "Strong Law of Small Numbers" [13]: each V_k is a mandatory exterior-calculus dimension $\binom{n}{p} = \dim \Lambda^p\mathbb{R}^n$ anchored in established physics before α^{-1} is computed, not an arbitrary selection from the small-integer pool. Under the topological-conjecture reading, the seven integers are quantized topological charges driving an extremal $\mathcal{N} = 2$ attractor mechanism [17] that fixes the moduli, and with them the IR EM coupling, to integer-determined values. **A laboratory precedent for "integer topology fixes a continuous coupling at 10^{-9} precision" already exists in the Quantum Hall Effect**: the integer-quantized Hall conductance is topologically protected by Chern-number invariance [25], fixing the von Klitzing constant $R_K = Z_0/(2\alpha)$ exactly. The framework's claim is structurally analogous: Pascal multiplicities act as topological charges at the attractor horizon, fixing α_0^{-1} at integer-determined precision. The framework's structural difference from Wyler's 1969 attempt [5] is decisive: Wyler's input was a continuous bounded-symmetric-domain volume that could drift with measurement refinement, whereas the $\{V_k\}$ are integer-quantized charges that cannot deform.

What remains open. A formal Lagrangian derivation is not yet established; Eq. (1) is stated as a topological conjecture (marked $\stackrel{?}{=}$ in the long companion). Because the Standard Model's $U(1)_Y$ sector is not asymptotically free (Landau pole), α_0 cannot be derived from the SM alone—a UV completion such as the M-theory cascade considered here is required. The candidate dynamical mechanism is iterated Atiyah-Bott-Berline-Vergne equivariant localization [22] on a 7-fold BPS-moduli-space filtration over a G_2-holonomy 7-manifold; under this framing the $\{V_k\}$ are equivariant Euler classes at fixed-point sectors and the cross-signature cascade labels reduce to fixed-point parametrization rather than continuous integration, by the same mechanism that yields closed forms in [23, 24]. Constructing the explicit prepotential is the principal open task.

Falsification criteria

The framework's α_0^{-1} is fixed by integer arithmetic and cannot drift continuously. Disconfirming observations: (i) confirmed multi-lab $|\dot{\alpha}_0/\alpha_0| > 0$ at $\geq 5\sigma$ (framework predicts $\dot{\alpha}_0 \equiv 0$); (ii) any canonically-anchored higher-dim object on Pascal rows ≤ 13 missed by the anti-diagonal (audit of all 105 entries plus a five-alternative-sequence null check at 15-digit precision returned zero candidates [27]); (iii) any first-principles derivation yielding cascade integers different from $\{28, 126, 210, 330, 165, 66, 13\}$.

CONCLUSION

A systematic survey of characteristic integers across mainstream and adjacent string-theoretic frameworks identifies a single contiguous anti-diagonal of Pascal's triangle whose seven entries $\{V_k\} = \{28, 126, 210, 330, 165, 66, 13\}$ are anchored independently of α^{-1} in established higher-dimensional physics. Their nested-reciprocal closed form (Eq. (1)) yields $\alpha^{-1}(0) = 137.0359990862878849\ldots$, matching CODATA 2018 at 0.11σ with seven integers and zero free parameters. Under the topological-conjecture reading, the integers are quantized topological charges fixing α_0^{-1} as a moduli-independent IR boundary condition, consistent with α_0^{-1} being a topological invariant of the M-theory IR vacuum. Falsifiable by confirmed

$\dot{\alpha}_0 \neq 0$ at $\geq 5\sigma$, or by any canonically-established higher-dimensional object on Pascal rows ≤ 13 missed by the anti-diagonal. The full derivation, the per-V_k identifications across the cascade, the matrix-theory algebra supplying the boundary $11^2 + 4^2 = 137$ reading, and the Pascal-uniqueness audit are given in the long companion [27].

The author thanks the Stanford Institute for Theoretical Physics (SITP) for early-stage exposure to these results. The full derivation, string-theory Pascal heat-map dataset, and verification code are archived in the long companion [27].

END MATTER

Verification code

The following Python computes Eq. (1) from the seven integers using exact rational arithmetic:

```
from fractions import Fraction as F
V = [28, 126, 210, 330, 165, 66, 13]
prods, p = [], 1
for v in V: p *= v; prods.append(p)
eps = sum(F(1, q) for q in prods)
alpha_inv = F(V[4] - V[0]) + eps
print(alpha_inv)
print(float(alpha_inv))
```

Output: 137.0359990862878849... (full reproducibility log archived in the long companion [27]).

Pascal-uniqueness audit

Of all anti-diagonal sequences of length 7 on Pascal rows ≤ 13, only $\{28, 126, 210, 330, 165, 66, 13\}$ has all seven entries individually anchored in established string-theory sectors as catalogued in Fig. 1, with $V_5 - V_1$ producing 137 as a Pascal entry within rows 7–14, and with the seven-term nested reciprocal sum reproducing CODATA α^{-1} at the part-per-billion level. The audit covered all 105 Pascal entries on rows ≤ 13. No alternative anti-diagonal satisfies all three criteria simultaneously. Full audit table in the long companion [27].

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